Matter is anything that has mass and takes up space. Matter is all of the liquids, gases, and solids around you. It is anything that is made up of atoms and molecules. Anything you can see or touch is made up of an atom. Atoms are too small to be seen without a microscope. All matter is the same, because all matter is made of atoms. The measurement of mass or other characteristics that can be seen without changing the way the object looks are called its physical properties. The most important physical properties are density and volume. Another way to study matter is by looking at chemical changes. The only way to look at chemical changes is to change the identity of the substance. Matter is always changing, and it happens right in front of our eyes.

Properties of Matter

Read the information and trace the dotted words .

Matter	anything that has mass and takes up space
Mass	the amount of matter in an object
Volume	the amount of space an object takes up
Weight	(on Earth) a measure of the force of gravity between Earth and an object
Density	a measure of how tightly packed matter is; the amount of mass contained in a given volume
Buoyancy	the upward push on an object by the liquid (or gas) the object is placed in
Conduct	allow heat or electricity to flow through
Insulate	not allow heat or electricity to readily flow through
Atom	the smallest unit of an element that still has the properties of the element
Element	a basic building block of matter; a pure substance that cannot be broken down into anything simple

Properties of Matter

Trace the dotted words.

Matter is all of the things that make up our environment. It is the air that we breathe, the water we drink, and the ground that we walk on. Objects that take up space and have mass are called matter. Mass is the amount of matter in an object. Mass is related to how much something weighs. Mass and weight are two totally different things. The mass is how much something weighs compared to other known masses. Weight is actually the force of gravity between Earth and the object. We use kilograms to measure an object's mass. To measure weight, scientists use a quantity called the Newton; one unit is 0.22 pounds. This helps scientists figure out the force of gravity between two masses. With the Newton measurement, scientists can figure out weight on any planet.

M_____ is anything that has mass and takes up space. Matter is all of the liquids, gases, and solids that make up the earth.

The amount of matter in an object is the object's m_____. Mass is reported in kilograms, and it is found by comparing other known mass amounts.

W_____ is different from mass because it measures the force of gravity between two objects.

The unit we use to quantify the difference in mass and weight is the N_____. Mass is the same everywhere in the universe. Weight changes because gravity is different. This is why we use the Newton to measure force.

Word Bank

matter	**weight**
mass	**Newton**

_____Matter is all of the liquids, air, and gases that make up the environment.

_____ The force of gravity between earth and an object is its mass.

_____ The Newton is the same as 0.44 pounds.

_____ Mass is reported in kilograms.

_____ The amount of matter in an object is the mass.

_____ Mass and weight are the same.

_____ If you use the Newton measurement, you can figure out the weight of a book in space and on the earth.

Matter and Mass

Write T or F in each blank.

_____Matter is all of the liquids, air, and gases that make up the
T
environment.

_____ The force of gravity between earth and an object is its
F
mass.

_____ The Newton is the same as 0.44 pounds.
F

_____ Mass is reported in kilograms.
T

_____ The amount of matter in an object is the mass.
T

_____ Mass and weight are the same.
F

_____ If you use the Newton measurement, you can figure out
T
the weight of a book in space and on the earth.

M	A	T	T	E	R	A	G	A	S
N	D	V	G	R	A	V	I	T	Y
Y	L	I	Q	U	I	D	T	H	W
K	H	Y	W	E	I	G	H	T	F
N	B	P	R	O	P	E	R	T	Y
E	R	W	R	S	V	M	A	S	S
W	W	R	L	O	M	B	Q	W	N
T	Q	K	I	L	O	G	R	A	M
O	X	N	R	I	F	D	P	G	H
N	R	T	S	D	N	M	W	T	E

Word Bank

weight	**liquid**	**Newton**
gas	**property**	**mass**
matter	**gravity**	**kilogram**
solid		

Matter and Mass

Find the words in the Word Search.

There are three important physical traits when dealing with matter. These are volume, density, and buoyancy. Volume measures how much space a sample of matter takes up. Scientists record volume in cubic centimeters. A sample of matter can be found by seeing how many cubes of a certain size you can fill.

Density is the measure of how tightly packed something is or the amount of mass contained in a given volume. Density helps explain how a huge metal boat can float. The boat is hollow and the air inside has very little mass but has a lot of volume. If all of this volume is filled up, it would become denser than water, thus sinking the ship. When any object is placed in water, the water pushes up on the object. If the push is strong enough compared to the objects weight, the object will float. This happens only when the object is less dense than the water. This upward push is called buoyancy. Objects have enough buoyancy to float when they are less dense than the liquid.

Read the information and trace the dotted words.

There are three physical traits when dealing with matter

v_____, d_____, and

b_____.

V_____ is found by seeing how many cubes

of a certain chosen size you can fill.

Volume is measured in c_____

c_____.

To measure how tightly packed matter is, you measure the objects

d_____.

B_____ is the upward push on

an object by the liquid the object is placed in.

Word Bank

volume	**buoyancy**
density	**cubic centimeters**

_____ can be found by seeing how many cubes of a certain size can be filled.

_____ is the amount of matter in an object.

The measurement of how tightly the matter in an object is packed is its _____.

The upward push of water on an object is called _____.

Volume is measured by _____.

The three most important physical properties of _____ are volume, mass, and density.

Word Bank

volume	**matter**	**density**
mass	**buoyancy**	**cubic centimeters**

_____ can be found by seeing
Volume

how many cubes of a certain size can be filled.

_____ is the amount of matter in an object.
Mass

The measurement of how tightly the matter in an object is packed

is its _____.
density

The upward push of water on an object is called

_____.
buoyancy

Volume is measured by

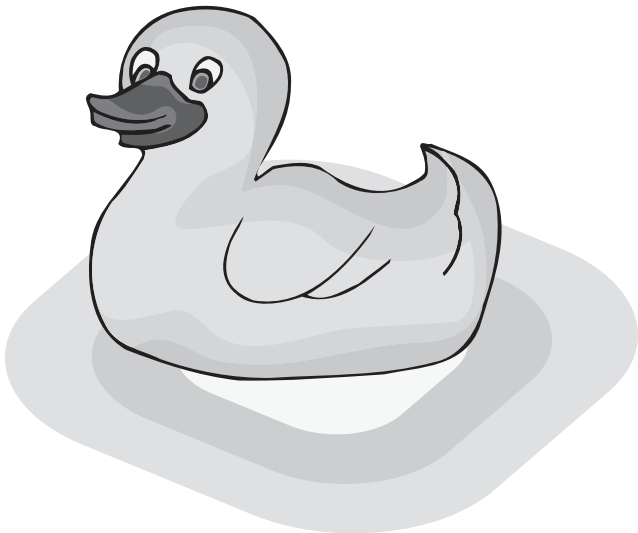
_____.
cubic centimeters

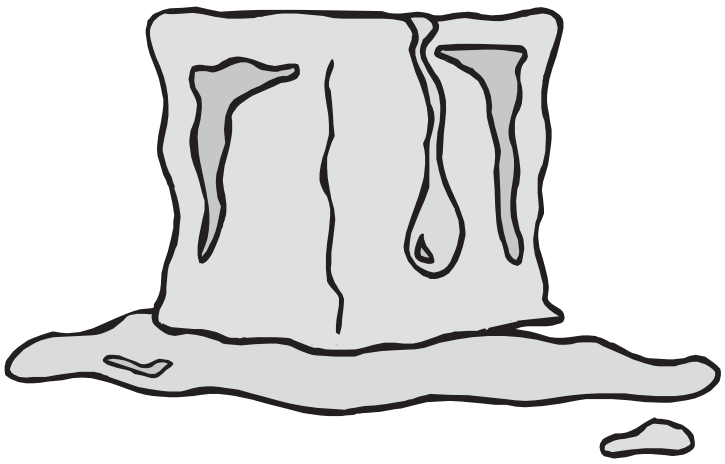
The three most important physical properties of

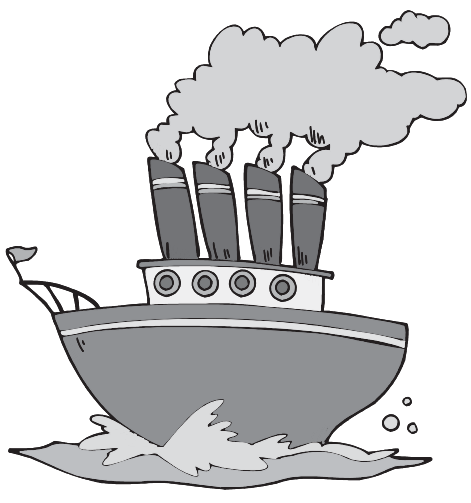
_____ are volume, mass, and density.
matter

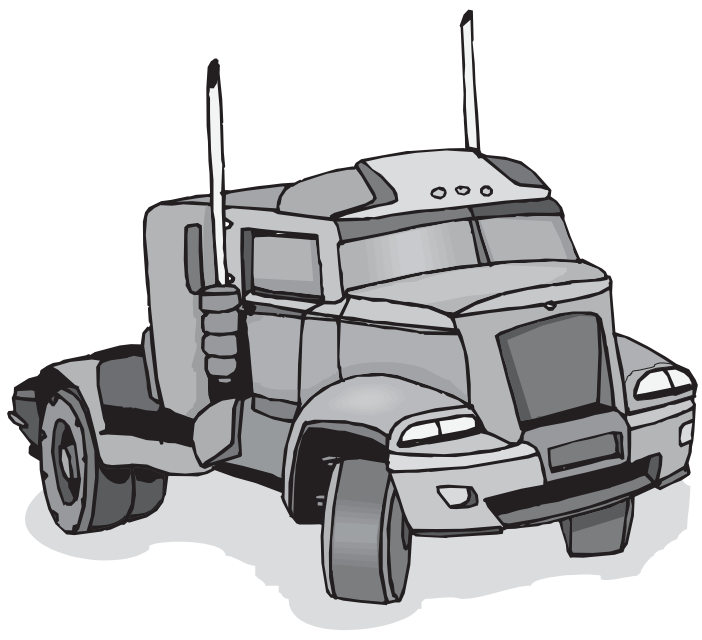
Word Bank

volume	**matter**	**density**
mass	**buoyancy**	**cubic centimeters**









Circle the objects that will float.

Matter has other important properties besides density. Some materials conduct heat or they insulate against the passage of heat. When a material is a conductor, it allows heat to flow through it readily. Some of the best conductors are copper, aluminum, and rock. When a material is used to insulate, it does not allow heat to pass through. In order for a conductor to work properly and be safe, it must also have an insulator. For instance, metal cooking pots are conductors, but they will always have a plastic or ceramic handle so you do not burn yourself. Insulation is also used to prevent noise and to keep your house warm or cool. Ceiling tiles are insulators in your school. They help soften the noise from the rooms around you. There is also insulation in your home. It is inside your walls and right above your ceiling. This type of insulation keeps your house cool in the summer and warm in the winter.

Conductors and Insulators

Read the information and trace the dotted words.

M_____ work as conductors and allow heat to pass through them.

The p_____ handle on a metal pot acts as an insulator to protect you from the heat.

Copper, aluminum, and rock are the most powerful c_____.

The tile on the ceiling of your school acts as an i_____ to soften the noise for the classrooms around you.

Word Bank

conductors	**metals**
insulator	**plastic**

Matter has other important properties besides density. It is also used to _____ and _____ heat.

_____ is one of the most powerful conductors. Copper, aluminum, and rocks are the conductors of the most _____.

Ceiling tiles in your school are used to soften the _____ coming from your classroom.

A _____ cooking pot must have a _____ handle to protect you from the heat.

Plastic _____ you by acting as an insulator.

Insulation is also used in your home to help control the _____.

Word Bank

conduct	**plastic**	**heat**
insulate	**metal**	**protects**
copper	**sound**	**temperature**

Conductors and Insulators

Fill in the blanks.

Matter has other important properties besides density. It is also used

to _____ and _____ heat.
conduct **insulate**

_____ is one of the most powerful conductors.
copper

Copper, aluminum, and rocks are the conductors of the most

_____.
heat

Ceiling tiles in your school are used to soften the

_____ coming from your classroom.
sound

A _____ cooking pot must have a
metal

_____ handle to protect you from the heat.
plastic

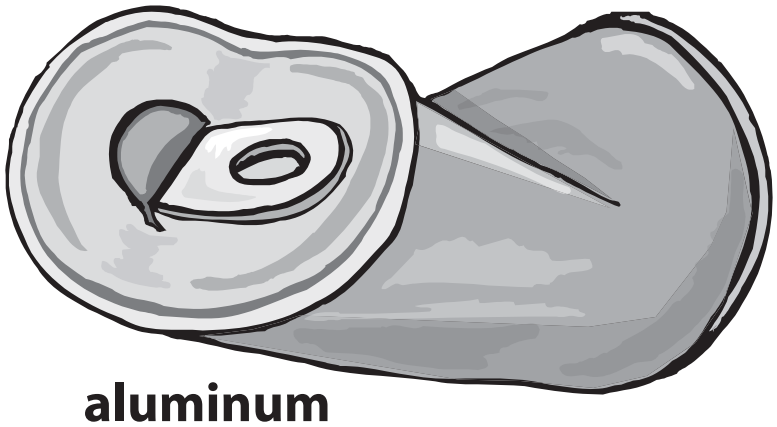
Plastic _____ you by acting as an insulator.
protects

Insulation is also used in your home to help control the

_____.
temperature

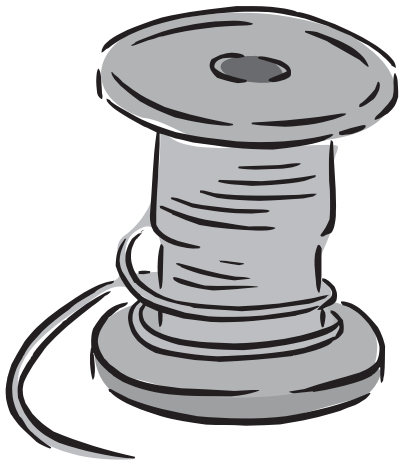
Word Bank

conduct	**plastic**	**heat**
insulate	**metal**	**protects**
copper	**sound**	**temperature**

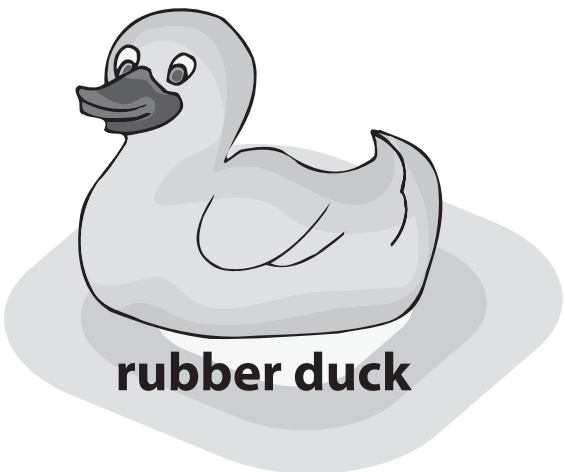


aluminum

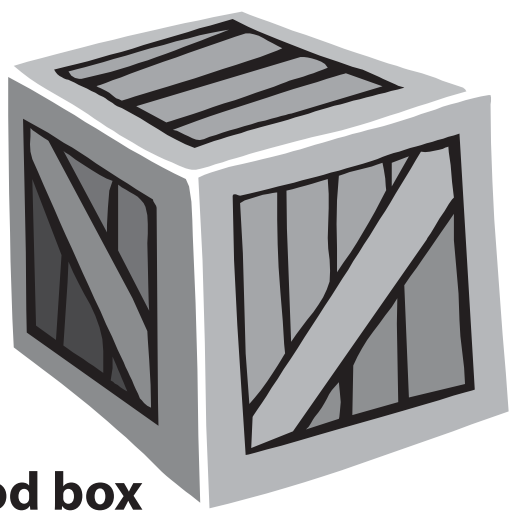
ceiling tile



copper wire

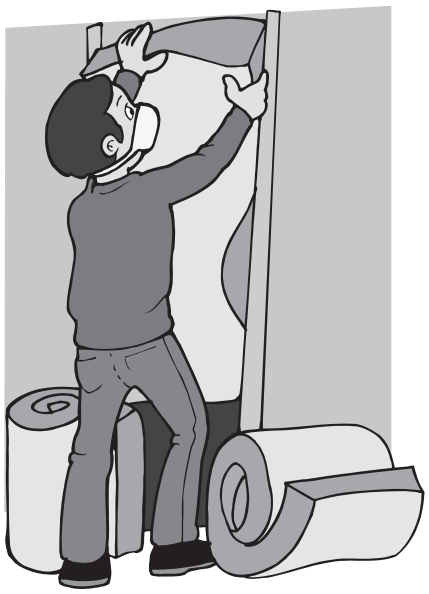


rubber duck



wood box

pots & pans



wall insulation

Conductors and Insulators

Circle the conductors and underline the insulators.

What is matter made of? Scientists cannot look inside of matter because the basic particles that make up matter are too small. In ancient Greece, Aristotle thought that all matter was made of four elements: earth, air, fire, and water. However, scientists have proven that matter is actually made of chemical elements. Elements are pure substances that cannot be broken down into smaller substances. A few of the elements you will know are gold (Au), aluminum (Al), carbon (C), & copper (Cu). Each of the elements is given a symbol. You can find the elements on the periodic table with each of their symbols. On the table, the metals are on the left, the non-metals are on the right and the metalloids are in the middle. The periodic table is a way to keep all of these elements organized.

What is matter made of?

Read the information and trace the dotted words.

Some of the basic building blocks of matter are

e_____.

The p_____ that make up matter are too small to be seen directly.

E_____ are a pure substance that can not be broken down into anything smaller.

Scientists organize these elements and their

s_____ on a chart called the

p_____ t_____.

Word Bank

elements	**symbols**
periodic table	**particles**

What is matter made of?

Fill in the blanks and trace the dotted words.

_____ The particles in matter are too large to be broken down.

_____ The symbol for gold can be found on the periodic table.

_____ We still believe that the four elements are earth, air, fire, and water.

_____ The periodic table helps scientists organize all of the elements in one place.

_____ We now know that matter is made of chemical elements.

_____ Each element is given a symbol of one or two letters.

What is matter made of?

Write T or F for each question.

_____ The particles in matter are too large to be broken down.

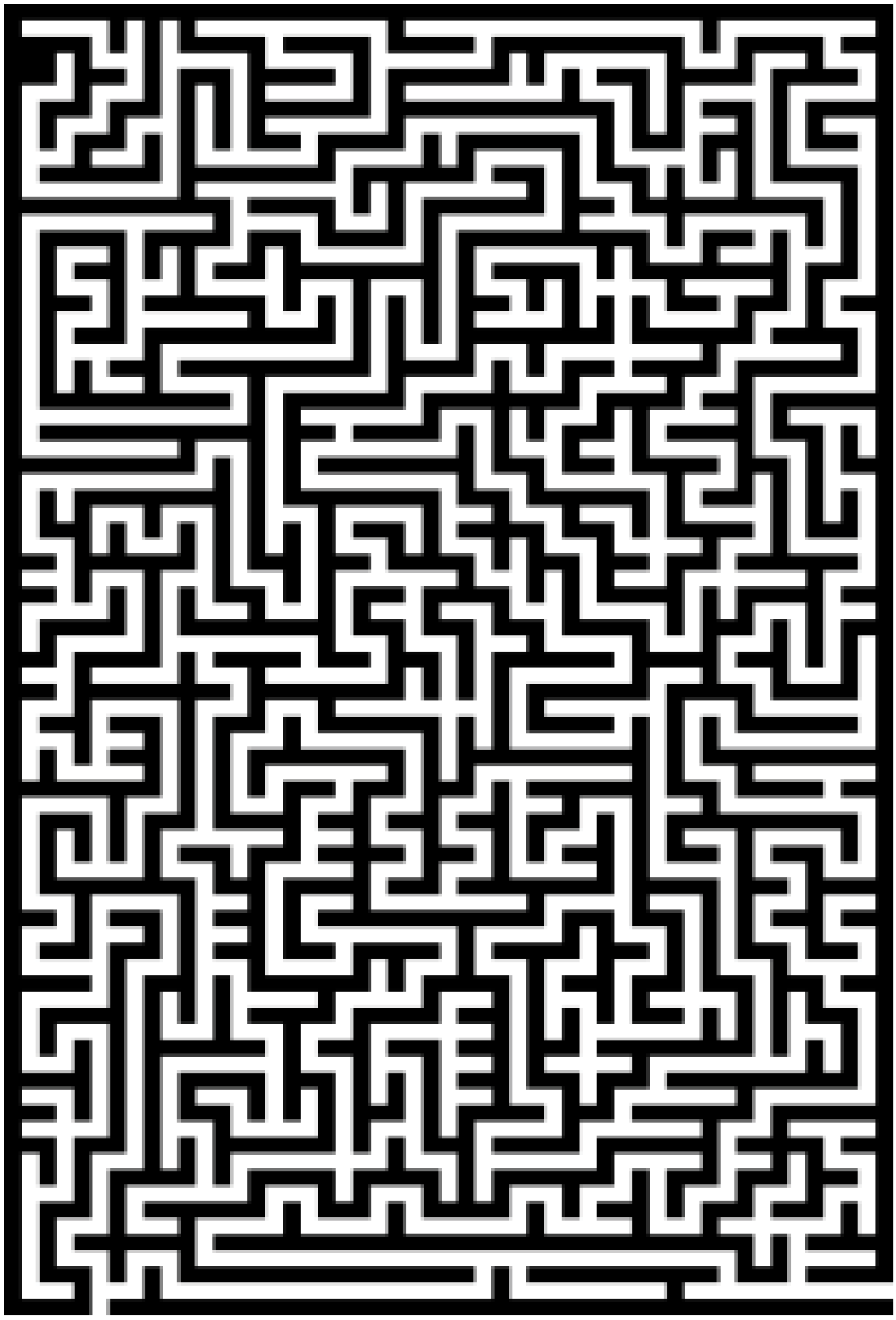
_____ The symbol for gold can be found on the periodic table.
T

_____ We still believe that the four elements are earth, air, fire, and water.
F

_____ The periodic table helps scientists organize all of the elements in one place.
T

_____ We now know that matter is made of chemical elements.
T

_____ Each element is given a symbol of one or two letters.
T



What is matter made of?

Help gold get to its symbol.

Elements are made of atoms. Atoms are the smallest unit of an element that retain the properties of the element. Atoms are so small that we can not even see them through a microscope. Scientists use a scanning tunneling microscope that can trace the bumps in a surface made by individual atoms. Now we know that matter is made up of elements and elements are made up of atoms. What are atoms made of? Atoms contain three particles called protons, neutrons, and electrons. The proton and neutron are located inside the nucleus, or center of the atom. The proton has a positive charge and the neutron is uncharged. The electron is located outside of the nucleus and has a negative charge. Protons carry one unit of positive charge and electrons carry one negative electric charge. Neutrons have no electric charge. All atoms carry equal amounts of protons and electrons, so they have no overall charge. The number of

What are elements and atoms made of?

Read the and trace the dotted words.

Scientists use a scanning tunneling microscope that uses a very sharp needle to trace the bumps in an a_____.

The p_____ is a positively charged particle and the number of p_____ determines what element the atom is.

The n_____ is located along with the proton inside the atom's nucleus.

The e_____ are located outside of the nucleus and they contain one negative electric charge.

Word Bank

atom	**neutrons**
proton	**electrons**

What are elements and atoms made of?

Fill in the blanks and trace the dotted words.

The smallest unit of an element that retains the properties of an element is an _____.

Using a regular_____ it is still impossible to see an atom.

The _____ and _____ are located inside the atom's nucleus.

The _____ has a _____ charge and is located outside of the atom's nucleus.

The _____ is the center of an atom.

Word Bank

protons	**positive**	**microscope**
negative	**electron**	**nucleus**
neutrons	**atom**	

What are elements and atoms made of?

Fill in the blanks.

The smallest unit of an element that retains the properties of an element is an _____.
atom

Using a regular_____ it is still impossible to see an atom.
microscope

The _____ and _____ are located inside the atom's nucleus.
protons **neutrons**

The _____ has a _____ charge and is located outside of the atom's nucleus.
electron **negative**

The _____ is the center of an atom.
nucleus

Word Bank

protons	**positive**	**microscope**
negative	**electron**	**nucleus**
neutrons	**atom**	

Example:

A small unit of element
Tiny so tiny
Only to see me with a microscope
Miniature

N _____

U _____

C _____

L _____

E _____

U

S

What are elements and atoms made of?

Write an acrostic poem with the word nucleus.

A compound is a substance formed when two or more elements are chemically joined. Water, salt, and sugar are examples of compounds. When the elements are joined, the atoms lose their individual properties and have different properties from the elements they are composed of. For example, water has the appearance of a single substance. However, it is made of hydrogen and oxygen. These two elements are chemically combined to make them act like a single substance. When hydrogen (H) joins oxygen (O), it has a new symbol called a chemical formula. Water is written as H_2O. A compound's chemical formula contains the symbols of the elements that make it up. The formula also contains numbers called subscripts. The subscript tells us the number of particles that combine together in a compound.

Compounds

Read the information and trace the dotted words.

A c_____ f_____ contains the symbols for the elements that make it up.

The c_____ f_____ for water is H_2O.

The number used in the chemical formula is the s_____.

When two elements are chemically combined it makes a c_____.

When compounds are chemically combined, the atoms lose their i_____ p_____.

Word Bank

compound	**chemical formula**
subscript	**individual properties**

Compounds

Fill in the blanks and trace the dotted words.

A chemical combination of two or more elements into a single substance is a _____.

When _____ and _____ are chemically combined, they make water.

Scientists write symbols for compounds called _____ _____.

The chemical formula for water is _____.

The number in a chemical formula is called a _____.

When _____ are combined, the atoms lose their _____ _____ and have different properties than the elements they are composed of.

Word Bank

compound	**subscript**	**individual properties**
oxygen	**elements**	**combine**
chemical formulas	**hydrogen**	**H_20**

Compounds

Fill in the blanks.

A chemical combination of two or more elements into a single substance is a _____.
compound

When _____ and _____
hydrogen **oxygen**

are chemically combined, they make water.

Scientists write symbols for compounds called

_____ _____.
chemical **formulas**

The chemical formula for water is _____.
H20

The number in a chemical formula is called a _____.
subscript

When _____ are combined, the atoms lose
elements

their _____
individual

_____ and have different
properties

properties than the elements they are composed of.

Word Bank

compound	**subscript**	**individual properties**
oxygen	**elements**	**combine**
chemical formulas	**hydrogen**	**H_20**

Word Bank

elements	**compounds**	**chemical formulas**
oxygen	**properties**	**subscript**
hydrogen	**individual properties**	

eygnxo _____

mecaihlc usfromal _____

dogehnyr _____

spocdounm _____

diiviladun potesreri _____

tmeelens _____

streieppor _____

prcsubtsi _____

Compounds

Unscramble the words.

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